

Ruturaj Jayant Kotwal

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SUMMARY

AI & software engineer with 3+ years of experience building cloud-native applications, data pipelines, and automation workflows across GCP, AWS, and modern JavaScript & Python ecosystems.

TECHNICAL SKILLS

- **Programming languages & frameworks** - Python, Javascript, Vue.js, FastAPI
- **Cloud** - AWS (Lambda, S3), GCP (BigQuery, Cloud Functions)
- **Machine learning & AI** - Gen AI (LLMs, VLMs, RAGs), Hugging Face, PyTorch, Tensorflow
- **Data analysis & visualization** - NumPy, Pandas, Matplotlib
- **System & tools** - Linux, Git, YouTrack

WORK EXPERIENCE

Thesis Student - Deutsche Bank AG | Jul 2025 - Sep 2025

- Designed an agentic-AI-driven automation framework for corporate action processing on Google Cloud.
- Built a proof-of-concept system by engineering BigQuery pipelines and prototyping autonomous workflows using Vertex AI.

Intern (TDI division) - Deutsche Bank AG | Jan 2025 - May 2025

- Completed the DB Cloud Developer sprint with hands-on development in GCP services.
- Supported production teams by debugging Python-based internal data-processing tools, improving system reliability.

Software Developer - Zoomifier Corporation | Nov 2021 - Feb 2024

- Migrated a legacy PHP CRM into a Vue.js component-based architecture, improving maintainability and user experience.
- Built event-driven AWS Lambda functions to process asynchronous API requests, reducing latency and enhancing scalability.
- Implemented multi-step automation workflows using Zapier to sync CRM data with multiple SaaS tools, cutting manual operations.
- Optimized API data-fetching logic, contributing to ~20% faster load times.

Software Development Engineer - Maverick Labs | Jun 2021 - Aug 2021

- Contributed to a JavaScript-based web application in an agile environment using Git and YouTrack.
- Developed and maintained Jest unit tests for UI components and utilities, improving code quality and stability.

Software Development Intern - Maverick Labs | Dec 2020 - May 2021

- Identified and resolved bugs in production code, gaining early experience in debugging, code reviews, and release cycles.

PROJECTS

Speech-controlled object recognition for Nao6 Robot

Tech Stack - Python, OpenCV, REST API

- Developed a Python and OpenCV solution for real-time voice-activated object recognition.
- Tested an AI-driven prototype that enabled responsive interaction through REST API integration.

E-Commerce performance & customer analytics platform

Tech Stack - GCP (BigQuery)

- Built a cloud data warehouse in BigQuery by processing and integrating 9 raw datasets (orders, payments, reviews).
- Delivered business insights through dashboards highlighting product performance, customer locations, and delivery–review correlations.

Machine learning for intrusion detection using CIC-IDS2017 dataset

Tech Stack - XGBoost, Keras

- Developed a high-accuracy (95%+) model to classify malicious vs. benign network traffic using the CIC-IDS2017 cybersecurity dataset.
- Leveraged XGBoost for intelligent feature selection and built a Keras-based MLP deep learning model for classification.

EDUCATION

M.Sc. Artificial Intelligence & Robotics | Hof University of Applied Sciences | Mar 2024 - present

M.Sc. Computer Science | Savitribai Phule Pune University | Jun 2019 - May 2021

LANGUAGES

English - C1 level | **German** - B1 level